SHIVARAMAN T

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Profile Summary

Electronics and Communication Engineering student specializing in AI and Cybernetics, with practical experience in Python-based signal analysis, sensor data acquisition, and embedded systems. Familiar with NumPy, Pandas, SciPy, Matplotlib, digital signal processing, FFT, Butterworth filtering, and peak detection. Interested in physiological signal quality, artifact investigation, waveform analysis, feature extraction, and reproducible biomedical data analysis.

Education

Vellore Institute of Technology Sep 2023 – May 2027
B.Tech. in Electronics and Communication Engineering (AI & Cybernetics) CGPA: 8.66/10

Technical Skills

Programming: Python, C, C++, Embedded C
Signal Processing: Digital Signal Processing (DSP), Time-Series Analysis, FFT, Frequency-Domain Analysis, Butterworth Filtering, Denoising, Peak Detection, Segmentation, Waveform Analysis, Feature Extraction
Python Libraries: NumPy, Pandas, SciPy, Matplotlib, scikit-learn
Biomedical Signals: PPG and ECG Analysis, Pulse Timing, Peak Matching, Statistical Analysis, Artifact Investigation, Signal Morphology Comparison
Embedded Systems: Sensor Interfacing, ADC Fundamentals, Data Acquisition, ESP32, Arduino, UART, I2C, SPI
Tools: Google Colab, Jupyter Notebook, MATLAB, Git, GitHub, LTspice, Cadence Allegro, OrCAD Capture CIS

Experience

Zettaone Technologies, Bengaluru Jun 2026 – Jul 2026
Hardware and PCB Research Intern

- Supported PCB research through circuit analysis, electronic design tools, and systematic hardware troubleshooting.
- Assisted with hardware verification, measurement workflows, and technical documentation to support design review.

Aqualis Inspection Services, Dubai May 2025 – Jun 2025
Engineering Intern

- Gained practical exposure to cathodic protection systems, inspection procedures, and engineering documentation.
- Supported inspection-related engineering tasks and documented observations for reliability-focused problem investigation.

Projects

PPG Signal Quality and Filtering Analysis GitHub

- Analyzed 5 PPG recordings sampled at 200 Hz using Python, NumPy, SciPy, and Matplotlib; reconstructed packet streams and identified a 200-sample gap in one recording.
- Applied an 8 Hz Butterworth low-pass filter and reduced first-difference standard deviation by approximately 85.9% in the evaluated noisy recording.
- Compared raw and filtered peak timing, matching 36 detected peaks within 150 ms and documenting signal-quality limitations and validation requirements.

ECG Signal Processing and Heart Rate Analysis GitHub

- Processed ECG time-series data to explore waveform characteristics and digital signal-processing techniques.
- Examined signal features and waveform patterns to support interpretation of cardiac activity.
- Explored peak-based beat timing and heart-rate-related measurements using signal-analysis methods.

Automatic Aerator System (AquaDox) GitHub

- Integrated dissolved oxygen, pH, and temperature sensors with a microcontroller-based data-acquisition system and Bluetooth communication.
- Implemented sensor interfacing and threshold-based control using Arduino, supporting automated environmental monitoring.
- Tested sensor readings and embedded-system behavior to support water-quality assessment and aeration control.

Publication

Smart Helmet Systems: IoT Enhanced Safety and Integrated Display for Two Wheelers
6th International Conference on Recent Advancements in Mechanical Engineering (ICRAME 2025), NIT Silchar Springer Link

Certifications

AWS Certified Cloud Practitioner | Signal Processing with MATLAB – Coursera | Arduino Programming and Embedded Systems – GUVI-HCL

Achievements

- Top 100 in the Zealestra X AWS Machine Learning Challenge among 5,000+ participants.
- Bharatiya Antariksh Hackathon 2024 participant.
- TCS CodeVita Season 13 – Global Rank 1347.